

PROBLEM STATEMENT

Square Root of Algebraic Sum

Find the derivative with respect to x of the function:

$$y = \sqrt{x + \frac{1}{x}}$$

STEP 1 প্রদত্ত অমূলদ ফাংশনটি উল্লেখ করি (Given Radical Function)

$$y = \sqrt{x + \frac{1}{x}} = (x + x^{-1})^{1/2}$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ করি (Differentiating both sides w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} \left[\sqrt{x + \frac{1}{x}} \right]$$

STEP 3 বর্গমূলীয় চেইন রুল প্রয়োগ করি $\left(\frac{d}{dx} [\sqrt{u}] = \frac{1}{2\sqrt{u}} \frac{du}{dx} \right)$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x + \frac{1}{x}}} \cdot \frac{d}{dx} \left(x + \frac{1}{x} \right)$$

STEP 4 অভ্যন্তরীণ রাশির অন্তরক নির্ণয় করি (Differentiating Inner Terms)

$$\frac{d}{dx} \left(x + \frac{1}{x} \right) = 1 - \frac{1}{x^2} = \frac{x^2 - 1}{x^2}$$

STEP 5 মান বসিয়ে বীজগাণিতিক গুণফল ও চূড়ান্ত সরল রূপ পাই (Final Algebraic Assembly)

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x + \frac{1}{x}}} \cdot \left(\frac{x^2 - 1}{x^2} \right) = \frac{x^2 - 1}{2x^2\sqrt{x + \frac{1}{x}}} = \frac{x^2 - 1}{2x^{3/2}\sqrt{x^2 + 1}}$$

FINAL ANSWER

$$\frac{dy}{dx} = \frac{x^2 - 1}{2x^2\sqrt{x + \frac{1}{x}}}$$

🔴 BOX 1: FORMULATION & RADICAL DIFFERENTIATION RULES

- **Formula:** $\frac{d}{dx} \sqrt{f(x)} = \frac{f'(x)}{2\sqrt{f(x)}}$.
- For $f(x) = x + x^{-1}$, $f'(x) = 1 - x^{-2} = \frac{x^2-1}{x^2}$.
- Simplification:

$$\frac{dy}{dx} = \frac{\frac{x^2-1}{x^2}}{2\sqrt{\frac{x^2+1}{x}}} = \frac{x^2-1}{2x^2} \cdot \frac{\sqrt{x}}{\sqrt{x^2+1}} = \frac{x^2-1}{2x^{3/2}\sqrt{x^2+1}}$$

⚠️ BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Derivative Sign of $\frac{1}{x}$:** Remember $\frac{d}{dx}(\frac{1}{x}) = -\frac{1}{x^2}$. Writing $+\frac{1}{x^2}$ is a catastrophic error.
- **Dropping the x^2 Denominator:** Writing $\frac{x^2-1}{2\sqrt{x+1/x}}$ without the factor of x^2 in the denominator.

🌐 BOX 3: REAL DOMAIN & CRITICAL VALUES

- **Real Domain:** We require $x + \frac{1}{x} \geq 0$ and $x \neq 0$.
- Since $\frac{x^2+1}{x} \geq 0 \iff x > 0$. Thus:

$$\text{Domain} = (0, \infty)$$

- **Critical Point:** $\frac{dy}{dx} = 0 \iff x^2 - 1 = 0 \implies x = 1$. At $x = 1$, $y = \sqrt{2}$ (global minimum).

💡 BOX 4: LEIBNIZ SUBSTITUTION (ALTERNATIVE VERIFICATION)

- Let $u = x + \frac{1}{x} \implies y = u^{1/2}$.
- $\frac{dy}{du} = \frac{1}{2\sqrt{u}}$.
- $\frac{du}{dx} = 1 - \frac{1}{x^2} = \frac{x^2-1}{x^2}$.
- By Chain Rule:

$$\frac{dy}{dx} = \frac{1}{2\sqrt{u}} \cdot \frac{x^2-1}{x^2} = \frac{x^2-1}{2x^2\sqrt{x+\frac{1}{x}}}$$

📊 BOX 5: AMIE EXAMINATION MARKS DISTRIBUTION (TOTAL: 3 / 4 MARKS)

- **1.0 Mark:** Correct chain rule setup for square root $\frac{1}{2\sqrt{x+1/x}}$.
- **1.0 Mark:** Correct differentiation of inner sum $1 - \frac{1}{x^2}$.
- **1.0 Mark:** Final algebraic simplification into unified fraction form.